

Who Will They Choose?: Avatar Selection Without a Gender-Race-Match

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Figure A



Figure B

Figure 1: A) Profiles of six avatars for users to choose between when asked to select the avatar they most identify with. B) The selection screen of the application used to investigate avatar choice.

ABSTRACT

Avatars in virtual spaces often serve as extensions of users. Avatar choice depends on context, with users typically selecting avatars they identify with. Race and gender can influence avatar selection. This study explores avatar choice when a gender-race-matched avatar is unavailable. We examine whether participants are more likely to choose a race-matched or gender-matched avatar in the absence of a gender-race-matched option. Results show that all participants select avatars matching at least one aspect of their identity. When forced to choose, participants favored gender-matched avatars over race-matched ones, suggesting gender may be more central to one's identity during avatar selection.

Index Terms: Avatars, Identity, Race, Gender

1 INTRODUCTION

Identity is shaped by various factors, including social categories like race and gender, which influence how individuals perceive themselves and others [6]. These identity traits even influence avatar choice during a virtual experience. One way users identify with their virtual avatar is by finding the similarities between themselves and their avatar [1]. In terms of race and gender, a user may identify with their avatar if the avatar has the same skin tone as they do or presents as the same gender. Users may also select an avatar based on how they anticipate an avatar with a specific identity would behave in a certain scenario [1]. For example, when asked to pick an avatar to complete a job interview, users may choose a female avatar over others due to gender stereotypes of competence and professionalism [7]. So, a user may choose an avatar they believe is most suited for a specific context or the avatar they personally identify with. However, without embedded customization, platforms can only include a discrete number of avatars of varying genders and races. Therefore, it is important to understand which

social categories guide avatar selection so that creators can make informed decisions about avatar diversity and understand the implications of the lack thereof.

As personal identity can influence avatar selection, when prompted to choose the avatar they most identify with in a video game or virtual reality application, what avatar does a user choose if a gender-race-matched avatar is not an option? In this work, we investigate whether participants will choose a gender-matched avatar, race-matched avatar, or an avatar with no matching identity traits when a gender-race-matched avatar is not provided as an option. Based on the institutional demographics where this study was completed, we chose to include female and male avatars who were White, Black, Hispanic, or Asian. The study and avatar selection was performed using a platform originally designed as a STEM educational tool. Due to the original purpose of this software, avatars that fit typical STEM stereotypes were intentionally omitted. Following the exclusion of White and Asian male avatars, six gender and race-diverse avatars remained for users to select.

We first hypothesize that when given the option, participants will choose the avatar that matches their self-reported gender and race. Given our pool of participants, the majority of participants were forced to choose an avatar with at least one identity trait different from their own. We aim to uncover if gender or race is more central to one's identity when choosing a virtual reality avatar to represent oneself. We infer that a majority of those without a gender-race-matched avatar will choose an avatar of their self-identified gender versus switching genders to choose a race-matched avatar.

2 METHODS

The experiment was approved by the Davidson College Human Subjects Institutional Review Board and was run at Davidson College. This research was completed in three CS courses, two of which were CS1 classes and the other was a 300-level Computer Graphics course. Participants were students from these courses and could drop out of the experiment at any time. Participants ($n = 20$) included 14 male and 6 female students. Students self-identified their race as 2 Black, 2 South Asian, 1 Middle East, and 15 White. 3 students identified as having Hispanic or Latin origin. Student ages were 17 – 23 years.

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During this study, students used a cross-platform smartphone-based VR application, developed in Unity. The Unity XR plugin was used to convert the application from a mobile device two-dimensional screen view to a stereoscopic view so that users saw the environment in three dimensions. The final application was downloaded on personal smartphone devices and placed into cardboard VR headsets. Smartphone devices were made available to students who did not have their own device.

When the study began, participants had no context for the scenario and were asked to choose the avatar they most identified with from a selection screen on the app, see Figure 1A. Avatars were characterized by varying skin tones and facial structures to depict different gender and racial groups, including a Hispanic female and male, a White female, a Black female and male, and an Asian female. These racial categories were included as they are the most common racial demographics of the population where the experiment was conducted. All avatars wore white lab coats to appear as scientists. After selecting an avatar, users placed their cellphones in the provided cardboard HMD and placed the HMD on their heads.

Once in the virtual reality experience, students found themselves in a virtual office and were co-located with their self-selected avatars. They were instructed, via written instruction in the application, to look around the room and find specific objects in the scene. After two minutes, as noted by the timer at the bottom right of the display, students removed the HMDs and completed an online questionnaire with questions related to their avatar selection, personal demographics, experience in the scene, belonging in Computer Science, and embodiment levels. We only analyzed the avatar choice and demographic data for this project.

3 RESULTS

Congruent with our first hypothesis, all participants chose an avatar that matched their self-reported race or gender. No participants chose an avatar they did not identify with through race or gender.

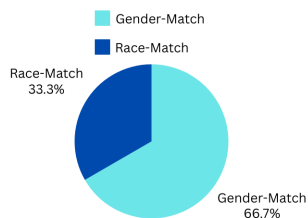


Figure 2: Avatar selection breakdown for participants without a gender-race-matched avatar available

The results also satisfied our second hypothesis. Of the 20 participants, 15 could not select an avatar that matched their race and gender. Among those without a gender-race-matched avatar, 67% chose a gender-matched avatar, while 33% chose a race-matched avatar. A single participant with a gender-race-matched avatar available chose an avatar of a different race. Four others were able to select a gender-race-matched avatar and chose that avatar.

4 DISCUSSION

When not provided an avatar that matches a participant's self-identified race and gender, three-fourths of participants selected an avatar of their same gender, but a different race. Leaving one-fourth who selected an avatar of a different gender but the same race. However, all participants chose an avatar with at least one matching identity trait. These findings suggest that gender, rather than race, plays a more central role in avatar selection, particularly when faced with limited avatar choices.

Since all participants chose an avatar that matched their self-reported race or gender, we can assume that social categories like race and gender influence identity and avatar selection. In virtual spaces, the avatar often represents the user in the virtual world. Therefore it follows that individuals may choose an avatar they identify with to "be" them. There are cases where users may choose an avatar they do not identify with, yet identity is still involved in this selection. In certain contexts, such as video games, users may choose an avatar they believe is stereotypically best suited for the scenario [4]. In other situations, people of color or women may choose to conceal their gender or racial identity given the culture of the space or presumed biases they may face [2, 3]. Since most of the individuals who were forced to choose an alternative avatar were white men, our research does not properly assess the avatar selection of women and people of color. Therefore, our findings may not be extendable to predicting which aspect of one's identity influences avatar selection more for people of color and women.

Given the demographics of our participants, a majority did not have the option to pick an avatar that matched their race and gender. Interestingly, when asked to choose the avatar they most identified with, 67% of those in this condition chose an avatar that matched their gender but not race. That leaves another 33% that chose an avatar with a different gender but the same race. Existing research supports that children see their gender as more central to their identity than their race [6]. This finding may explain this pattern in avatar choice. The context of the virtual reality application may have also influenced avatar choice. Given the avatars wear lab coats like scientists, and individuals tend to choose an avatar they believe is stereotypically best fitted for the scenario [4], male participants may be more likely to choose a male avatar of a different race instead of a female avatar of the same race due to masculine stereotypes of STEM spaces. Further research should be completed that explores whether or not the same trends in avatar selection appear in a variety of virtual reality contexts.

Another implication of our work includes the biases that may be associated with our findings. Time spent in a black avatar has been shown to reduce racial bias for lighter-skinned participants [5]. Therefore, an extension of this research may investigate if avatar choice exposes internal biases, and time spent in a contrasting avatar may reduce these biases. Altogether, this work expands our understanding of user interaction with avatars and the influence of identity in virtual experiences.

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